

## CHAPTER 2: SOFTWARE DESIGN

<Brief description of chapter 2>

### 2.1 Description of the Design Solution

#### 2.1.1 General Description

This section provides a general perspective/overview of the entire design solution.

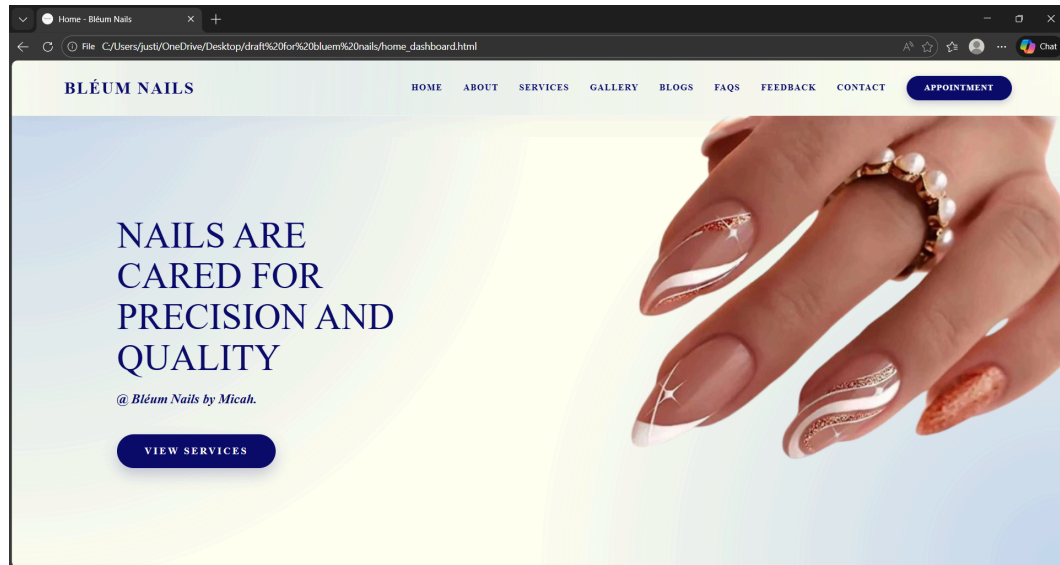


Figure xx Illustrative Diagram of the System

#### 2.1.2 Engineering Principles Involved

In developing our program, we used principles such as modularity, abstraction, and iterative refinement. These were the important steps in the initial stage of defining user needs, system components, and data flow that led to the application of the KISS (Keep It Simple, Stupid) principle, better requirement analysis, and at the same time, clarifying the project's vision from the very beginning. (GeeksforGeeks, 2025) This upfront planning minimized risks and aligned development with user needs. The elements that the principle of reusability and test-driven development (TDD) reflect, which we used to create highly independent, therefore, testable modules.

Throughout implementation, we continuously improved the website through peer reviews, revised the codes, and received input thereby, our work was in alignment with the practices of continuous integration that along with the SOLID principles ensured the maintainability and scalability of the product. The frontend part was concentrated on cleanbuilding of the HTML/CSS/JavaScript for maximum UX, while the backend was oriented towards secure data handling and efficiency. Besides minimizing the number of errors, this cyclical mode of working also made the product ready for future improvements.

#### Engineering Principle

Iterative software development studies provide solid evidence for its effectiveness in projects like ours. They show that delivery capability improves and defects get detected early, through data-driven analysis of open-source repositories. A 2018 empirical study published in IEEE Transactions on Software Engineering looked at iterations from five large projects. They used feature aggregation and graph-based measures to accurately predict project outcomes, surpassing the baselines in error metrics, thus proving that iterative refinement plays a significant role in risk reduction and quality enhancement.

Besides that, research into broader principles such as upfront planning and SOLID also indirectly supports the draft-first approach, which is to some extent reflected in empirical comparisons of SDLC models, where iterative methods are definitely more flexible and can easily adapt to changes through incremental builds. A Semantic Scholar paper on agile iterative development linked it to higher quality of web applications through surveys, the emphasis being on continuous improvement which significantly enhances the security and efficiency aspects.

### **2.1.3 Prior Art Analysis**

These systems provide calendar-based slot inventories, pricing, and promotion rules configured by the provider, and marketplace-style business discovery. Various other patents for on-demand salon or mobile barber services mainly focus on mechanisms to route customer requests to the nearest available stylist, the integration of calendaring widgets in websites, the implementation of service delivery optimization for efficient scaling. Personnel supply chain booking patents take this concept further by proposing a centralized database that coordinates various providers and resources.

This project allows a single admin the advantage of simply and easily be able to manually override the date/price. This type of administrative control is quite different from what is already existing in the market, in which most of them are rule-based, data-driven discounting and enterprise-level settings. In addition, your automatic PDF receipt generation with GCash integration is focusing on localizing payment handling to Philippine e-wallet norms. At the same time, these patented and commercial systems explain pricing and promotional instruments in a more abstract way without referring to local wallets. Lastly, by incorporating native watermarked gallery uploads directly to the admin dashboard, you are embedding the intellectual property protection into the system's core. On the other hand, booking and salon tools nowadays are more concerned about availability, pricing, and logistics rather than incorporating built-in watermarking of the creative work.

Table xx Prior Art Analysis Matrix

| Design           | Features              |                        |                       |                  |              |               |
|------------------|-----------------------|------------------------|-----------------------|------------------|--------------|---------------|
|                  | Online Booking System | Automated PDF Receipts | Client Design Uploads | Allergy Tracking | Cash Payment | Gcash support |
| Nails by Ann     | X                     |                        |                       |                  | X            |               |
| Nail Garden      | X                     |                        |                       |                  | X            |               |
| Mich & Myl Nails |                       |                        |                       |                  | X            |               |
| PROJECT          | X                     | X                      | X                     | X                | X            | X             |

**Legend:**

**X = Feature Present**

**(Blank) = Feature Not Found or Not Available.**

## 2.2 General System Architecture

This section describes how the engineering concepts and elements for the nail services website are implemented to create a functional solution for the business.

### 2.2.1 Hardware Elements

**Client Devices:** The system is designed to be accessed via smartphones and personal computers (PCs) by customers looking to book services.

**Administrator Hardware:** The business owner uses a computer to access the administrative dashboard, manage the gallery, and track payments.

**Development Machine:** A laptop is utilized for the coding, testing, and debugging of the web application.

### 2.2.2 Software Elements

#### A. Embedded Software

Not applicable, as the system is entirely web-based and does not require specialized hardware firmware.

#### B. Application Software

**Web Frameworks and Languages:** The system is built using HTML and CSS to establish a minimalist and elegant user interface.

**User Interface (UI):** The UI features a mobile-friendly design that adapts to various screen sizes and includes a toggle for dark and light modes.

**Admin Dashboard:** A specialized interface allows the administrator to view analytics, upload watermarked photos to the gallery, and manually adjust service prices.

**Booking Subsystem:** A calendar-integrated module handles appointment scheduling and automated reminders for clients.

#### C. Key Algorithms Used (Optional)

**Capacity Limiting Algorithm:** A logic filter is implemented to ensure the system only accommodates a maximum of three clients per day.

### 2.2.2 System Algorithm

The main system algorithm manages the flow of a booking from start to finish:

1. **Availability Check:** The system checks the database for available dates manually set by the admin.
2. **Booking Validation:** It verifies that the daily limit of three clients has not been reached before confirming a new slot.
3. **Payment Processing:** Upon service completion, the algorithm tracks whether the payment was made via Cash or GCash.

### 2.2.3 Data, Datasets, and Processing

#### a. Datasets

- **Generated Data:** The system uses mock data to represent client profiles, including names, phone numbers, and potential allergies.
- **Service Data:** A static dataset contains the list of available services (e.g., Gel overlay, BIAB, Soft gel extensions) and their fixed prices.

## **b. Data Processing Scheme and Algorithms**

- **Input Validation:** Raw data from client inputs, such as "preferred nail designs" or "feedback," is processed to ensure it meets the required format before being saved.
- **Gallery Processing:** Admin-uploaded images are processed to include watermarks before they are displayed in the public gallery.

## **c. Other Data Utilized in the Design**

- **Service Constants:** The system stores constant values for pricing, such as ₱199 for Gel overlays and ₱499 for Soft gel extensions.
- **Client Records:** A database of client-specific information, such as appointment history and rebooking preferences, is maintained for personalized service.

## **2.3 Design Alternatives**

### **2.3.1 Rationale for Design Alternatives**

The development of the Bléum Nails By Micah Appointment System is proposed to address the inefficiencies associated with the current manual appointment scheduling process used by the system. Schedule conflicts, delayed responses, and faulty record-keeping might result from the current method of managing appointments, which involves handwritten diaries and chat platforms. Customer satisfaction and service delivery may suffer as a result of these problems. By giving customers real-time access to service availability, the proposed system seeks to streamline the appointment booking process and make it easier for them to make appointments. In order to decrease no-show issues and enhance overall operational efficiency, the system will also arrange client records and apply structured payment verification through GCash deposits.

Additionally, the suggested solution would give a centralised digital platform that improves the home-based business's professional image by providing integrated social media links for improved client engagement and transparent service listings. To guarantee usability, dependability, security, and maintainability, the system will be designed in compliance with ISO/IEC 25010 software quality standards and under the direction of OWASP security principles. In order to promote the long-term scalability and sustainability of the business operations, a mobile-first responsive design strategy will also be used to guarantee accessibility for users who predominantly access services through smartphones.

### **2.3.1 Design Alternative 1:**

**Traditional multi-page web architecture** design focuses on a classic server-side rendering approach, prioritizing simplicity and search engine visibility.

## **A. Engineering Principles of Alternative**

**Server-Side Rendering (SSR):** This principle involves generating the full HTML for a page on the server in response to a request. This reduces the processing burden on the client's device (Fielding, 2000).

**Relational Data Modeling:** Utilizing **Third Normal Form (3NF)** for the database schema to ensure data integrity and minimize redundancy in appointment records (Codd, 1970).

**Session-Based Authentication:** Utilizing server-side cookies to manage administrative sessions, providing a stateful and secure connection for the salon owner.

## **B. Architecture of Design Alternative**

A monolithic architecture is used in the system's construction. The server retrieves the available time slots from the SQL database, inserts them into an HTML template, and then returns the finished page to the client when they request the booking page. CRUD (Create, Read, Update, Delete) activities that are directly connected to server routes are used in the administrator dashboard.

## **C. Constraints**

**Constraint A: Usability:** High for desktop users, but may feel inefficient on mobile due to constant page refreshes.

**Constraint B: Performance:** Slower "per-click" speed as every interaction requires a full page reload from the server.

**Constraint C: Reliability:** Highly reliable; if the server is up, the logic is guaranteed to execute.

**Constraint D: Maintainability:** Very high (Simple Code Score); the codebase is easy to follow for any standard web developer.

## **iv. Evaluation Results (if model)**

### **2.3.2 Design Alternative B**

**Single Page Application (SPA)** design prioritizes a "seamless" user experience, making the website feel like a fluid mobile application.

## **A. Engineering Principles of Alternative**

**Asynchronous JavaScript and XML (AJAX):** Allows the web page to update parts of the UI without reloading the whole page, improving the perceived speed (Garrett, 2005).

**Component-Based Architecture:** The UI is broken down into independent, reusable pieces (for example is Calendar Component, or Service Card), making the interface highly modular.

**Client-Side Routing:** Navigating between the Homepage and Booking sections happens instantly within the browser, managed by a JavaScript router.

## **B. Architecture of Design Alternative**

The architecture of the system is decoupled. When a client selects a date, the front-end sends a JSON request to the back-end, which only provides the required information. The back-end is a RESTful API, while the front-end is a static bundle of files. After that, the UI changes dynamically.

### **C. Constraints**

**Constraint A: Usability:** Extremely high and the fluid transitions provide a modern, "app-like" feel that users prefer.

**Constraint B: Performance:** Fastest interaction speed after the initial load, though the very first visit may take slightly longer to load the JS bundle.

**Constraint C: Reliability:** Average and if the user has a momentary internet "hiccup" or what we call delay or lag, the JavaScript state might need a manual refresh.

**Constraint D: Maintainability:** Medium (Moderate Complexity); requires managing both a front-end and a back-end codebase.

## **2.3.3 Design Alternative C**

**Progressive Web App (PWA) with Cloud Integration** design focuses on mobility and offline resilience for the home-based technician.

### **A. Engineering Principles of Alternative**

**Service Workers:** Scripts that run in the background, separate from the web page, enabling features like offline access and push notifications (Russell & Gaunt, 2017).

**Optimistic UI Updates:** The system visually confirms an action like (for example: uploading a receipt) immediately, then syncs with the server in the background to mask network latency.

**NoSQL Data Persistence:** Using a document-oriented database to handle flexible data structures, such as varying nail service descriptions and image metadata.

### **B. Architecture of Design Alternative**

The system makes use of a PWA manifest and Serverless Functions. It enables the owner of the salon to "install" the website on the home screen of their phone. Because IndexedDB is used to cache the data locally on the device, the owner can access the daily schedule even if internet connectivity is patchy.

### C. Constraints

**Constraint A: Usability:** Best for mobile-first users; provides the ability to access the schedule without a browser address bar.

**Constraint B: Performance:** Excellent; utilizes local caching to serve pages almost instantly.

**Constraint C: Highest;** the "Offline-First" approach ensures the schedule is accessible even without a stable connection.

**Constraint D: Maintainability:** Lower (High Complexity); managing service worker lifecycles and cache synchronization increases code complexity.

### 2.4 Standards Involved in the Design

This section presents the standards followed by the design, including their references. Matrix may be used to show how standards are used in each specific design.

Table xx Summary of Standards Involved in the Alternatives

| Standard         | Brief Description                                              | DESIGNS                                          |          |          |
|------------------|----------------------------------------------------------------|--------------------------------------------------|----------|----------|
|                  |                                                                | DESIGN A                                         | DESIGN B | DESIGN C |
| ISO/IEC 25010    | Systems and software Quality Requirements and Evaluation       | Used for performance and maintainability checks  |          |          |
| W3C HTML5 / CSS3 | World Wide Web Consortium standards for modern web development | Applied for standard desktop and mobile layouts. |          |          |



|                      |                                                                       |    |    |                                             |
|----------------------|-----------------------------------------------------------------------|----|----|---------------------------------------------|
| OWASP ASVS           | Standard for testing and verifying web application security controls. | NA | NA | Focused on secure login and authentication. |
| iCalendar (RFC 5545) | Standard for representing calendar events and scheduling data.        | NA | NA | Used for simple appointment scheduling      |

The table above presents the mapping of specific engineering standards to the three design alternatives. This matrix is used to justify the technical robustness of each design based on industry-recognized benchmarks:

- **Quality and Maintainability:** **ISO/IEC 25010** is applied specifically to all design to ensure that the system meets strict performance and maintainability checks. This ensures the software remains reliable under consistent use by the salon staff.
- **Web Architecture:** All front-end development for **all designs** adheres to **W3C HTML5 / CSS3** standards to guarantee cross-browser compatibility and a responsive mobile-first layout.
- **System Security:** **Design C** prioritizes high-level security by implementing the **OWASP Application Security Verification Standard (ASVS)**. This focuses on secure login, authentication, and the protection of sensitive client data such as contact details and payment records.
- **Scheduling Interoperability:** **Design C** also incorporates the **iCalendar (RFC 5545)** standard to handle appointment scheduling. This allows the salon's booking system to be interoperable with external calendar applications, facilitating easier reminder generation for clients.

In summary, the integration of these engineering standards serves as a foundational framework for the design and development of the online booking platform. By adhering to **ISO/IEC 25010**, **W3C**, **OWASP**, and **iCalendar** protocols, Team Sentinails ensures that the final product is not only a functional scheduling tool but a secure, reliable, and high-quality software solution. These standards collectively bridge the gap between manual booking and automated efficiency, fulfilling both the client's needs and professional engineering requirements.